INCH-POUND
MIL-PRF-3098/76E
12 January 2005
SUPERSEDING
MIL-PRF-3098/76D
27 August 1997

PERFORMANCE SPECIFICATION SHEET

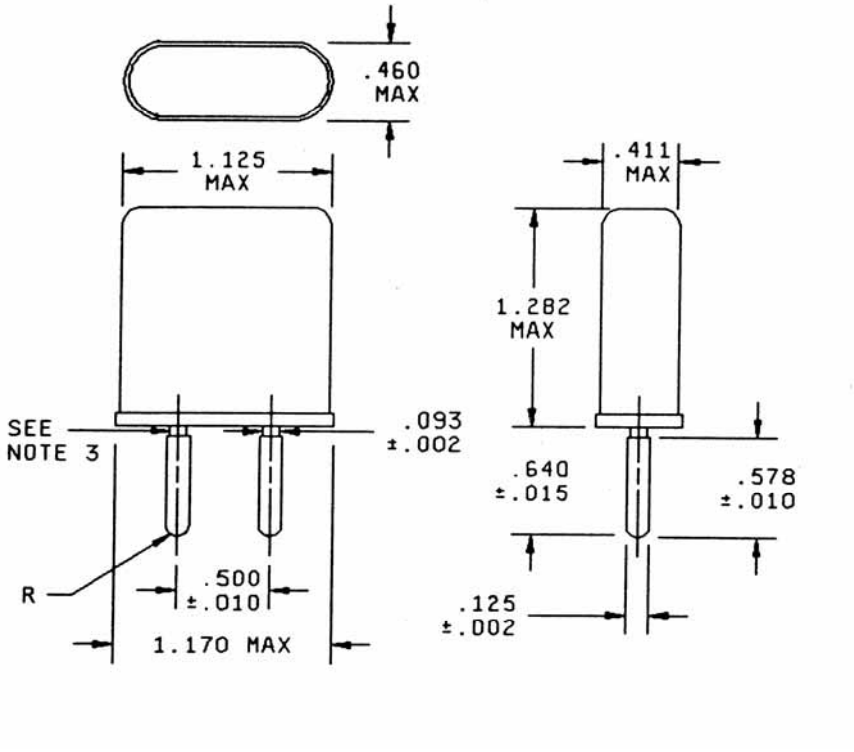
CRYSTAL UNIT, QUARTZ, CR1/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 3.6 MHz to 10 MHz; fundamental; noncontrolled; antiresonance.

Inches	mm
.002	0.05
.010	0.25
.015	0.38
.093	2.36
.125	3.18
.411	10.44
.460	11.68
.500	12.70
.578	14.68
.640	16.26
1.125	28.58
1.170	29.72
1.282	32.56



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR1/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 3.6 MHz to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Antiresonance, load capacitance: 35.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
3.6 to 5.5	90
5.5+ to 6.5	60
6.5+ to 7.8	50
7.8+ to 10.0	30

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0794)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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